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The Expanding Realm of Music Therapy

Music therapy has experienced significant growth worldwide in recent years. As global uncertainties, migration, political divisions, and terrorism increase, life becomes more challenging. Music, renowned for its positive impact and lack of side effects, is increasingly used to alleviate stress caused by modern lifestyles.

Music resonates with everyone capable of hearing, from newborns to the terminally ill, and from the physically robust to the impaired. Expanding research confirms that long-term musical engagement enhances cognitive abilities, including linguistic skills, reasoning, and creativity, fostering social adjustment and peace. Music stimulates the brain; playing instruments, for example, engages vision, hearing, touch, motor planning, emotion, and symbolic interpretation, activating diverse brain regions. Notably, some Alzheimer's patients retain musical memory long after losing other recollections.

At a fundamental level, musical forms stimulate adaptation, deeply rooted in our autonomic nervous system. These connections between life processes and music remain despite illness or disability, independent of musical skill. Consequently, music therapy effectively addresses emotional, cognitive, and developmental needs stemming from learning difficulties, mental and physical ailments, abuse, stress, and terminal illnesses.

Music therapy distinguishes itself by its reliance on music, an integral part of human existence since our origins. It has served as a mental companion, from ancient people facing nature's wrath to modern individuals confronting societal pressures. Even science, often skeptical of abstract concepts, acknowledges music's power. Extensive research validates its therapeutic and prophylactic roles, leading to its acceptance as a complementary medicine for conditions like epilepsy, mental disorders, speech impairments, and terminal illnesses. Patients report experiencing a renewed sense of dignity, finding comfort and peace, through music therapy.

Despite its familiarity, music therapy remains complex. This text explores its multifaceted dimensions, often defying simple analysis. The patterns in music, mirroring those in nature, contribute to its subjective nature. Ethnomusicology highlights music's role in binding and energizing communities, aiding them in overcoming adversity.

While musical engagement has been prevalent for centuries, its therapeutic potential was rediscovered during World War II. Observing positive physical and emotional responses in wounded soldiers, hospitals began hiring trained musicians.

Today, music therapy is recognized as most effective when integrated with other interventions. Sessions often employ diverse techniques, such as imitation, accompaniment, dialoguing, and rhythmic grounding, requiring flexibility from the therapist.

A music therapist designs, conducts, evaluates, and documents therapeutic sessions. This involves gathering patient background, conducting assessments to identify strengths and weaknesses, and formulating realistic goals. Assessments focus on how clients create, respond to, and relate to music. Treatment plans outline strategies and musical experiences tailored to client needs. Institutional settings often involve interdisciplinary collaboration. After a period of therapy, progress is evaluated,

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and adjustments are made as needed. Upon completion, therapists guide clients in self-music therapy.

Beyond therapy sessions, therapists may develop listening programs, organize musical events, and collaborate with educators. They may also consult with other professionals and manage administrative tasks.

A typical therapist's routine includes:

- Assessing individual and group needs.
- Planning and implementing music sessions.
- Documenting and evaluating progress.

Modern music therapy employs various approaches, tailored to individual needs through trial and error. While specific guidelines vary, regular sessions are recommended, ideally lasting one to two hours with intervals. Unlike pharmaceuticals, music therapy poses no risk of harmful side effects.

Is music necessary for human existence?

Scientific evidence increasingly confirms music's fundamental role in human life. From ancient nomadic, tribal, hunter-gatherer, and settled societies, music has been a constant presence. Recent findings highlight the potential of music therapy to enhance cognitive abilities. Psychologists now recognize the benefits of pleasant music for individuals affected by traumatic experiences. Growing evidence underscores music's importance in improving cognitive function, particularly in premature infants. Furthermore, numerous positive reports indicate significant behavioral improvements in individuals with intellectual disabilities through music training. Confirmed accounts also suggest that music helps people regain lost confidence and cope with various limitations.

While Mozart's music has often featured prominently in modern experiments, diverse musical forms, including Indian ragas, are also proving to be therapeutic for those experiencing stress and distress. Researchers observing music's undeniable positive emotional impact on distressed individuals are finding scientific validation. Ample evidence suggests music elevates mood in stressed and depressed individuals by promoting brainwaves associated with relaxed alertness. Clinical trials, notably those involving Mozart's music by scientists like Alfred Tomatis, Frances Rauscher, and Gordon Shaw, consistently point to music's positive influence on overall human health and well-being.

Although the therapeutic potential of music has long been acknowledged, it has only recently been subjected to rigorous scientific scrutiny. Despite its celebrated status in legends and myths, music's impact was largely considered subjective. However, recent decades have witnessed a surge in scientific understanding of music's effects on the mind and body, largely due to advancements in neuro-imaging techniques, leading to the emergence of 'neuromusicology.' While this field is still in its early stages and further research is needed to fully understand music's preventative and therapeutic mechanisms, the direction is clear. In the future, music could move beyond being seen as alternative medicine and become a respected complementary therapy alongside established medical fields like anesthesia, geriatrics, physiotherapy, psychotherapy, rehabilitation, and surgery.

Despite its lack of guarantees, music's significance as a companion for the ill, lonely, and marginalized is well-established. It fosters meaningful communication, especially for those without support or companionship. A healthy and positive outlook on life is within reach for anyone who listens. Music

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provides reliable support during feelings of helplessness, depression, and despair. It acts as a moral booster, revitalizing the mind and bringing joy. In this context, Aaron Copland's profound words resonate: "The greatest moments of the human spirit may be deduced from the greatest moments in music!"

Recognition and Perception of Music

Scientific reports indicate that while basic auditory perception might have innate components, the recognition of sound patterns, particularly in music, is largely a learned process. Research in Psychology Today (Oct 2000) highlights music's complex neural representation, making its study challenging. A key aspect of music perception is the context-dependent meaning of notes within a melody, as demonstrated by studies where animals showed varied neuronal responses to the same tone in different melodic contexts. This reveals that neurons learn to prioritize sounds based on their significance, such as a tone signaling food, leading to increased cellular responses. This finding revolutionized the understanding of brain organization, indicating that learning occurs not just in higher cognitive centers but also within the sensory systems themselves, shaping how we interpret and assign meaning to the intricate patterns of sound.

Ecstasy through Music

Witnessing audiences' ecstatic reactions to musicians in musical evenings highlights a phenomenon that neuro-physiologists define as an altered state of consciousness resulting from integrated brain activity. Ancient traditions have long recognized that achieving a state of ecstatic bliss (ananda) is possible through the unification and transcendence of cognitive and emotional processes. This ecstatic state is believed to be beneficial in combating increasing feelings of alienation and disconnection, as well as in processing trauma. Historically, yogis have employed sound vibrations, specifically selected sounds and rhythms, alongside consistent and dedicated practice (sadhana) often involving intense visual stimuli, to attain such states. Notably, these diverse practices are reported to share underlying physiological mechanisms. Recent research from Mt Sinai has revealed that acoustic brainwave entrainment can alter brainwave patterns, leading to synchronization in the theta (3-6 cps) to slow alpha (6-8 cps) range, facilitated by the development of serotonergic circuits connecting the limbic brain to lower brain regions. Remarkably, these induced brainwave changes have the potential to influence fundamental aspects of an individual's emotions, self-perception, and identity, factors that often contribute to human suffering.

What is Music Therapy?

Music therapy, grounded in the mind's associative and cognitive capabilities, is increasingly recognized as a complementary or integrative medicine. This expressive therapy offers promising potential for accelerating and enhancing patient treatment when combined with conventional medical interventions like medication, surgery, or rest. Typically administered by qualified practitioners or sometimes by patients themselves, music therapy involves the clinically prescribed use of music, with or without specific therapeutic goals, aimed at fostering overall positive changes across cognitive, physical, psychological, and social domains. As adaptable and creative as music itself, it provides a spontaneous means to harness music's appeal for individuals of all ages and abilities. Kenneth Bruscia (1989) defined music therapy as a "systematic process of intervention wherein the therapist helps the client to achieve health, using musical experiences and the

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relationships that develop through them as dynamic forces of change," identifying systematic intervention, a therapeutic relationship, and musical experience as essential elements.

The foundation of music therapy lies in the understanding that the capacity for musical self-expression and experience is an inherent right. However, social barriers such as tyranny, socio-economic disparities, fear of authority, and feelings of shame can stifle emotional expression, leading to cumulative negative impacts on mental and physical well-being. In this context, music plays a crucial role by providing limitless avenues for the release of suppressed feelings like disappointment, helplessness, or frustration. Remarkably, this outlet often remains accessible even in cases of accident, disability, injury, or illness. Therapeutic music is inclusive, with both modern and traditional forms demonstrating beneficial effects. Even individuals in hospice care can find noticeable relief through listening to or expressing their pain through music, offering a sense of peace and pleasant anticipation. Ultimately, music therapy fosters contentment and confidence, fundamental requirements for a fulfilling life.

As a form of complementary medicine, music therapy aligns with a growing trend in healthcare. Complementary medicine refers to healing strategies used alongside conventional medical practices, rather than as standalone treatments. This integration of complementary therapies, including music therapy, into mainstream medical care has led to the emergence of "integrated medicine," a model where mainstream and complementary approaches are combined within a practice or institution.

Music Therapy: Down the Ages

Throughout human history, music has profoundly influenced both individual and social evolution, serving as a potent tool for expression due to its extensive range and depth of sound frequencies capable of resonating deeply without causing harm. Intriguingly, even expressions of painful emotions in music often evoke pleasant sensations, a quality that prompted the poet Shelley to note, "The sweetest songs are those that tell of saddest thoughts." This characteristic explains why musicians may experience pleasure even while performing melancholic pieces. While the term "music therapy" gained prominence during the World Wars, the application of music for its healing properties is an ancient concept.

Archaeological evidence reveals the consistent use of music, primarily in song and rhythm, across virtually all past human settlements and cultures. The discovery of playable flutes over 9000 years old in China and bone fragments with drilled holes at Neanderthal sites suggests the early and widespread presence of musical instruments. Long before European acoustics, ancient civilizations in Arabia, Greece, China, and India were well-versed in the preventative and therapeutic roles of musical sounds and their vibrations. Greek legends spoke of music healing Ulysses' wounds, and the Arabian writer Ibn Sina documented music's therapeutic effects in his medical treatises. Ancient Chinese civilization viewed rites and music as instruments for maintaining a harmonic balance between Heaven and Earth, a concept still relevant today for its cathartic, healing, and educative potential.

Great thinkers of the past, including Aristotle and Plato, recognized music's capacity to alleviate human suffering. Al Farabi, a 9th-century Turko-Persian musicologist and psychologist, extensively discussed music's therapeutic effects on the soul in his work "Meanings of the Intellect." Robert Burton, in his 17th-century "The Anatomy of Melancholy," highlighted the importance of music and dance in treating mental ailments, particularly melancholia. Charles Darwin also explored the potential co-evolution of music and human civilization.

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The systematic and regular use of rhythmic or tonal vibrations for healing has been integral to ancient cultures and traditions, evident in systems like tantrism, shamanism, and nada yoga, often aligning with religious practices in Buddhism, Hinduism, Islam, Sikhism, and Sufism. These religions incorporated the repetition of powerful tones, rhythms, pitches, and other sound vibrations in their daily rituals to facilitate deeper states of consciousness. Hazrat Inayat Khan, a Sufi who taught classical music in the early 1920s, observed that the entire body, including muscles, blood circulation, and nerves, is moved by the power of vibration, stating that the human body is a "living resonance of sound," with sound affecting every cell, gland, circulation, and pulsation—an observation increasingly supported by modern therapists. Tibetan monks utilize the singing bowl with its significant harmonic overtones.

The evolution of music reveals a reliance on melodies in primitive forms worldwide. The introduction of sophisticated instruments like the piano led to a shift from just intonation to equal temperament scales, creating new sonic experiences. Primitive societies maximized music's use, integrating rhythms and melodies into all human and social activities, from agriculture and hunting to simple household tasks and celebrations. Music was also employed in mourning rituals, such as the "oppari" in Tamil Nadu, which, while expressing sorrow, aids in overcoming loss and healing emotional wounds. Thus, even negative emotions conveyed through music can touch listeners without causing harm, instead fostering healing and resilience. For both the primitive person fearing nature and the modern individual fearing their own species, music holds immense power. Reintroducing the musical culture of primitive societies into modern activities like computer work, workshops, and offices could enhance meaning and foster camaraderie by reducing tension and anxiety.

What does Music Therapy Promise?

While music therapy doesn't claim to directly cure specific ailments, music therapists believe its promise lies in fostering robust immune mechanisms and the potential to restore the body's healthy functioning, ultimately leading to an improved quality of life. Notably, unlike many other therapies, music therapy uniquely addresses a patient's emotional, physical, and spiritual needs simultaneously.

The modern era's competitive lifestyle, with its excessive focus on performance and results, is a significant source of stress across all demographics, including students, teenagers, adults, executives, politicians, and seniors. Individuals from various professions, such as students, company employees, law enforcement, government workers, medical professionals (especially surgeons), and airline staff (particularly pilots and flight attendants), are all susceptible to stress in different ways. This stress can manifest as irritability and anger, leading to negative interpersonal interactions that further exacerbate the problem in a vicious cycle. Externally induced stressors, whether physiological or emotionally threatening, disrupt the body's equilibrium.

It is now widely understood that an individual's psychological and physiological response to ongoing stress places immense strain on themselves and their surroundings. This strain can manifest as heightened irritability, insomnia, lethargy, nervousness, and dizziness, as well as physical symptoms like neck and muscle stiffness, lower back pain, vision problems, appetite loss, acidity, indigestion, irregular bowel movements, vertigo, and nightmares.

Coping with stress, psychologists now recognize, hinges on an individual's evaluation of the stressors. Recent research suggests that the stressor itself is not inherently stressful; rather, the individual's perception of it determines the severity of the problem. This is where music proves particularly useful. With its boundless patterns and nuances derived from endless combinations of tonal frequencies and rhythmic consistencies, music helps to alleviate the rigidity or gravity associated

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with stressors. Furthermore, music enables individuals to effortlessly perceive stressors in a more positive and realistic light, drawing upon their own learning and experiences, and promotes a state of relaxed alertness conducive to understanding and addressing potential stressors.

Music Therapy: The Probable Mechanism

Many therapists believe that the "enjoyment factor" or "feel good factor" inherent in music is key to its therapeutic efficacy. While music's mechanisms remained largely enigmatic until recently, advances in human brain research, particularly with sophisticated electronic instruments, have significantly illuminated our understanding. It is now widely accepted that various emotional disorders and tension-induced outbursts can be effectively managed by carefully selecting and administering appropriate musical pieces, including assertive musical phrases, melodies, harmonies, and songs (with or without lyrics). Such auditory interventions have proven helpful in calming feelings of rage or anger. The use of ancient and traditional musical instruments like the didgeridoo, damru, conch shell, ek tara, bells, and singing bowls has also considerably contributed to a pacifying effect on restless or agitated minds.

A remarkable aspect of music therapy is that diverse musical experiences can positively address a wide range of a patient's health or immune needs. Because emotions in music are conveyed within pleasant settings and patterns, even expressions of anger or violence lack bitterness and are presented in a positive and agreeable manner. The anger, sorrow, and conflict we encounter in music are perceived as pleasant, distinguishing them from the often unpleasant and sometimes traumatic emotions experienced in non-musical (real-life) contexts. Consequently, when a patient in a sorrowful state listens to a melancholic melody, the inherent pleasantness within the music, when synchronized with their own sorrowful mood, transforms the latter into a more bearable experience. In essence, music enables individuals to cope with otherwise unpleasant situations by converting them into something more agreeable. Through music, negative emotions like sorrow, bitterness, frustration, and anger can be positively framed, becoming more pleasant and manageable.

Personal Tempo of the Patient/Client

El-Sheikh, a music therapy researcher, proposes that music exerts a scientifically measurable effect on individuals based on their unique "personal tempo." This personal tempo is defined as the rate of a person's spontaneous auditory, visual, speech, sensory relational, and response behaviors in relation to the music tempo they experience. In music, "tempo" refers to the time interval between two beats.

According to El-Sheikh, music can have three distinct effects on human behavior depending on its tempo relative to an individual's personal tempo. Music with a tempo matching one's personal tempo is believed to stimulate the right hemisphere of the brain, responsible for holistic functioning, including simultaneous processing of multi-sensory input for a comprehensive understanding of the environment, such as visual-spatial skills. This stimulation can induce a temporary state of forgetfulness, offering a respite from life's difficulties.

Conversely, music with a tempo faster than the listener's personal tempo is suggested to stimulate the left hemisphere of the brain, impacting reasoning and potentially creating tension, stress, and fatigue in the outer nerve receptors, which could lead to hallucination and partial dreaming. In contrast, a song tempo that stimulates both hemispheres of the brain is associated with relaxation and pleasure while maintaining awareness of one's surroundings.

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Regarding the role of nostalgia in music preference, Abdel-Salhim, another music therapist, posits that it depends on an individual's relationship with the tempo of a musical piece. He suggests that a soothing tune, upon inducing ecstasy, prompts the mind to develop pleasant memories associated with that feeling, such as the image of a loved one or the sound of birdsong. Abdel-Salhim argues that it is not the happy memories themselves that soothe us, but rather the specific music tempo that elicits this effect.

Some Methods and Strategies

Music therapists work with diverse populations possessing varied needs. Their musical expertise equips them with the skills to discern suitable rhythms and resonance capable of assisting individuals in overcoming challenges by redirecting their focus away from the source of their pain. Through a range of musical exercises and strategies, refined by past experiences, therapists design sessions and select appropriate music tailored to the client's specific treatment plan.

A Typical Therapeutic Setting

In contemporary therapeutic settings, music activities are strategically employed to achieve various benefits, including composing songs for emotional expression, playing instruments like the piano to enhance fine motor skills, utilizing instruments for nonverbal emotional improvisation, improving patient movement, and fostering communication with individuals with intellectual disabilities.

In American hospitals, music therapists adapt to the diverse needs of numerous patients, often using a mobile cart equipped with instruments, songbooks, and recordings. They conduct sessions ranging from 15 to 60 minutes per patient, adjusting to individual responses and convenience. Initial visits involve assessing the patient's musical preferences. Subsequent sessions may incorporate singing, playing simple rhythm instruments like drums, learning instruments such as the guitar, QChord, or keyboard, listening to live vocal or instrumental music performed by the therapist, engaging in relaxation exercises with background music, and composing original songs. Family members and visitors are often encouraged to participate, enhancing patient involvement. Sessions can involve the therapist providing music or collaborative music creation between the therapist and patient.

Music therapy sessions can be conducted in group or individual settings, typically lasting 30 to 60 minutes with a frequency of one to five times per week. A common opening in US sessions is a lively "hi song," followed by interventions tailored to individual goals and concluding with a summarizing "bye" song. These interventions, prescribed by the therapist, may include movement to music, music games for social interaction, songwriting, lyric analysis and discussion, singing and vocal improvisation, playing instruments and instrumental improvisation, music and relaxation activities, music and guided imagery, music performances, and drum circles. Core activities encompass creating original music (improvising or writing lyrics), guided imagery and relaxation, listening to live and recorded music, movement, playing instruments, singing, and vibrotactile experiences. It is crucial to recognize that a standardized session is ineffective; therapists must design sessions and select music specifically for each patient's treatment plan. In a hospital ward, therapists manage the needs of multiple patients sequentially, moving between rooms with their musical equipment and assessing each patient's preferences before initiating music.

Wheeler's (wheel chair bound patient) supportive activity-oriented music therapy utilizes group singing, instrument playing, or musical games to promote healthy behavior and participation among

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disoriented, withdrawn, or anxious clients. These activities aim to improve social interaction and awareness, maintain reality orientation, divert from obsessions, encourage successful group involvement, manage impulsive behavior, and foster healthy leisure time use. In contrast, catharsis-oriented music therapy employs musical activities to address subconscious conflicts and traumas, providing a means of self-expression for suppressed feelings, thoughts, and memories. Interestingly, group therapy focused on releasing bottled-up emotions often yields significant benefits within the group, enhancing social interaction.

Indian music shows considerable promise in music therapy, as various ragas are believed to reflect different emotions and nuances, capable of inducing desired mood changes. Through selected ragas, clients can experience and identify emotions, facilitating the understanding of others' emotions and aiding in the modulation of one's own emotional experiences.

Neurophysiological research in the late 1980s by Thomas M. McKenna, using melodies with varying contours but the same five tones, demonstrated that single neuron responses in cat auditory cortices varied with the melodic contour. Neuronal firing depended on a tone's position within a melody, with cells responding differently when a tone was preceded by others versus being the first. Furthermore, the same tone elicited different reactions depending on whether it was part of an ascending, descending, or more complex contour. These findings underscore that the pattern of a melody is crucial, indicating that auditory processing is not a simple transmission of sound.

Simple Strategy for Relaxing

Simple relaxation can be achieved by following these steps:

- Find a calm and private space, such as a living room, garden, or shaded area in a playground, where you are unlikely to be interrupted.
- Dim most of the lights, leaving only a night lamp on.
- Stretch your legs out or sit in a comfortable lotus posture, or simply find the most comfortable seated or lying position on a mattress (being careful not to fall asleep).
- Close your eyes gently.
- Take several deep breaths, allowing your entire being to harmonize with the environment.
- Smile and cultivate a feeling of love for everything within and around you, without any judgment.
- Play soft, peaceful, and soothing music, focusing your attention on it and minimizing any external distractions.
- Begin a process of relaxation from your head down to your toes, repeatedly assuring yourself that you will remain relaxed, trusting that the music will aid you in this process.
- Listen to the music with love and a smile... Relax!

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Designing Music Therapy sessions

Contemporary music therapists integrate traditional healing methods like chants and vibrations with modern innovations to serve an expanding range of patient populations. In recent years, they have developed diverse and innovative music exercises, strategies, and interventions tailored to the specific needs of their clients.

Music therapy sessions are professionally designed, considering the various factors influencing an individual's health and well-being. A primary focus for therapists is selecting the most appropriate musical inputs to elicit a therapeutic impact. This involves choosing from a vast repertoire, with the selection process heavily influenced by the client's psychological and emotional state, age, lifestyle, and past experiences (both positive and negative). These factors help the therapist understand the client's condition.

Once the therapeutic goals are established based on the client's needs, the therapist designs music therapy sessions to engage the client in various musical experiences using specific methods and techniques. In this process, the therapist must carefully select the types of music and musical experiences most relevant to the client. There are six fundamental categories of musical experiences employed in therapy: chanting, listening, re-creating, composition, improvisation, and envisioning. While a performer conveys existing musical ideas, an improviser or composer works with their own creative expressions within the therapeutic context.

Chanting

Chanting, an age-old practice prevalent across cultures and religions, serves as a time-tested method for overcoming grief and trauma. It involves the repetitive recitation of either nonsensical phrases or assertive, self-affirming resolutions, aiming to alter consciousness by quieting the troubled mind and, for some, to enhance psychic energy.

Various religious traditions incorporate chanting. Hinduism and Buddhism utilize the syllable "Om," representing the universal consciousness (Brahman). Islam involves chanting the "Ninety-Nine Beautiful Names" of Allah. Christianity includes the chanting of Jesus' name in prayer, alongside distinct Western chants (Gregorian, Gallican, Mozarabic, Ambrosian) and Eastern forms (Byzantine, Syrian, Armenian, as well as Coptic and Ethiopian chants). Jewish chants typically draw from Biblical texts. Notably, Dr. Alfred Tomatis considered Gregorian chant an excellent auditory stimulus for focused attention, potentially accelerating learning by naturally containing the sound elements necessary for a state of "Mind alert/Body relaxed."

Many music psychotherapists view chanting as a beneficial psychoacoustic process for inducing relaxation. Some psychologists suggest that the mental habituation resulting from chanting might stem from auditory pattern recognition, eventually leading to passive hearing, which could be the key to the relaxation response.

Interestingly, the Gothic preference for high harmonics, characterized by relatively bright sounds rich in upper partials, utilizing the higher vocal register and clean melodic transitions, is recognized as a soothing experience.

In Eastern traditions, Tibetan singing bowls and gongs, crafted from an alloy of seven metals, produce similar effects. Monks often visualize these instruments as closed lotus buds, with the petals unfolding one by one as the bowl is sounded. These bowls are traditionally used for physical healing in hypnotic states, serving as a conduit for transporting consciousness through the body and

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envisioning biochemical changes that promote health and vitality. They are believed to help restore the body's original harmonic frequencies, consistently eliciting positive responses in listeners. Sometimes, a bowl is placed on the body and sounded, allowing vibrations to spread internally like a massage. To harmonize the chakras through sympathetic resonance, the bowl can be placed over chakra points and sounded. Surrounding a person with multiple sounding bowls creates a virtual "sound bath."

A typical singing bowl produces between two and five distinct tones. Sound waves resonate within the bowl while also radiating outwards. The overtones within the fundamental tone create an initial perception that unfolds for the listener. The "beat" frequencies found in high-quality bowls result from the simultaneous passage of two or more wave trains through the same area, potentially synchronizing brain hemispheres with a binaural beat and encouraging entrainment. Through focused concentration, one can attain a brainwave frequency in the alpha range (14-17 Hz), a state of "relaxed alertness" where body and mind merge with retained consciousness and self-control.

The singing bowl can proactively support meditative techniques, aiding in the release of bodily and mental tension and enhancing awareness of the energy currents within and around us. Sensitivity to these energies is believed to increase with practice.

Similarly, the unique sound vibrations produced within our own bodies, known as overtones, have a soothing effect on the mind and behavior. Often described as "sounds within a sound," overtones represent a series of harmonic frequencies related to the fundamental pitch of a note or voice. Buddhist monks visualize overtones as the petals of a lotus bud gradually unfolding into full bloom, with the ultimate experience being the appreciation of the bloom's beauty, complexity, and delicacy through its unfolding.

'Aum' Chanting

Aum (also known as Om or, less commonly, Ohm), represented by the symbol ॐ, is a mystical and sacred syllable in various Indian religions, including Hinduism, Jainism, Buddhism, and Sikhism. It is frequently used to commence religious texts and conclude recitations.

Despite diverse religious affiliations, Aum serves as a reliable mantra for all who chant. A simple yet effective method for chanting "Aum" involves the following steps:

1. Select a short phrase, such as "Aum."
2. As you chant (intone) the phrase, gradually slow down the articulation.
3. While intoning, direct your focus to the exhaled breath. Visualize the subtle flow of air within your body that accompanies the chanting. Starting from the abdomen, trace the air's passage into your mouth cavity, moving through the stomach, heart, and throat. Pay particular attention to the movements occurring in your larynx and mouth, and be sensitive to the qualities (such as warmth or texture) of this outgoing breath.
4. Simultaneously, as you intone, focus your attention on the natural resonators within your body, such as your mouth, throat, sinuses, windpipe, and lungs. Identify their shapes and internal structure.
5. Allow yourself time to listen to the unique "music" that resonates within you.
6. Cultivate the practice of an "inner smile," which will naturally manifest externally.

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The unique sound vibrations generated within our own bodies, known as overtones, have been observed to produce a soothing effect on the mind and behavior.

The Ways of Chanting the Vedas: An Illustration

In India, Vedas are the most ancient of treatises, which have survived for over 5000 years. Its association with tones and melodies have enabled their easy memorization. It is known that the Vedas were chanted in eight different ways. At present, however, only five have survived.

They are :

1. Vakya or samhita : Also referred to as mula (Base), this method refers to enunciation of the full sentence.
2. Pada : In this method, each word in a sentence is uttered.
3. Krama: Here, the words are used in certain fixed patterns. For instance, the pattern of (1-2), (2-3), (3-4), (4-5) etc.
4. Jatha : Here the pattern of chanting is : (1-2, 2-1, 1-2), (2-3, 3-2, 2-3), (3-4, 4-3, 3-4) etc.
5. Ghana: Here the pattern of chanting is : (1-2, 2-1, 1-2-3, 3-2-1, 1-2-3), (2-3, 3-2, 2-3-4, 4-3-2, 2-3-4) etc.

Illustration of the chanting pattern of a vedic text:

Namah Shambhave Cha Mayobhave Cha Namah ShankarayaCha
Mayaskaraya Cha Namah Shivaya Cha Shivataraaya Cha

(Meaning : I bow to Shiva, the creator of auspicious, delightful and beneficial events and deeds; I bow to the one who is exceedingly auspicious).

Now one can find how this text is chanted:

Namah Shambhave (1-2)

Shambhave Namah (2-1)

Namah Shambhave Cha (1-2-3)

Cha Shambhave Namah (3-2-1)

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Gopal Park, Behind Image Sonography Clinic,
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Namah Shambhave Cha (1-2-3)

Shambhave Cha (2-3)

Cha Shambhave (3-2)

Shambhav Cha Mayobhave (2-3-4)

Mayobhave Cha Shambhave (4-3-2)

Shambhave Cha Mayobhave (2-3-4)

Listening

Listening to live or recorded music, while seemingly the simplest music therapy procedure, can be surprisingly complex. While many find stress relief in listening to "good" music, individual taste and preference are paramount. Forcing a client to listen to relaxation music they dislike can be counterproductive, potentially increasing rather than decreasing stress.

Listening differs from other musical experiences as it primarily involves receiving music without active participation in its creation. However, the listener remains actively engaged in a way distinct from improvisers, composers, and performers, who have direct creative responsibilities.

Music listening is beneficial for clients needing activation and for calming tense or agitated individuals, both children and adults. It often fosters a more softened outlook on life. Hospitalized patients, for instance, commonly find music listening helpful for relaxation and stress reduction. It can also aid in pain management and regulating bodily functions like heart rate and breathing. Depending on the music's genre and structure, it can also be arousing and energizing. For psychiatric patients, listening to songs can bring deep-seated emotions to the surface, facilitating expression and sharing. When combined with psychotherapy, music listening can stimulate images, fantasies, associations, and memories, significantly enriching the therapeutic process. It also assists the elderly in structured reminiscence and life review.

Perceptual listening tasks are employed with children who have learning disabilities or intellectual disabilities, helping to develop their auditory processing skills. Listening to songs and following lyrics can aid in learning and memorizing colors, numbers, vocabulary, behavioral sequences, and various academic subjects that might otherwise be challenging or unappealing. A common experience for many is the mood enhancement that music listening brings, sometimes leading to profound peak experiences. These experiences can instill hope, courage, and a renewed appreciation for life's beauty, while also fostering confidence and optimism. Listening can be adapted to be "active as well as passive" or "physical as well as intellectual," leading to a state music psychologists term "relaxed alertness"—a state of being both alert and relaxed, often achievable through music without the need for drugs or other external aids.

While music listening forms a significant part of the therapeutic process, complementary therapies such as standard exercises, meditation, and relaxation techniques are also often recommended, particularly in de-addiction programs.

Re-creating Music

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Musical recreation in therapy harnesses the power of imagination and creativity. The inherent rhythms and melodies of music offer boundless opportunities for imaginative exploration, which can be both enjoyable and relaxing. Music therapists utilize re-creative activities to redirect their patients' or clients' minds away from worry and sorrow-filled thoughts. This approach is often prescribed for individuals needing to develop sensory motor skills and learn adaptive behaviors. It is believed to aid in maintaining reality orientation and in mastering various social roles.

Re-creating music helps to instill positive, healthy, and feel-good resonance and rhythms into the thought processes of participants. These activities have been found to relax troubled or traumatized minds, effectively refreshing and positively impacting mental states. Furthermore, the re-creative musical structure can foster the development of independent self-expression and build confidence in both performers and listeners.

Given its inherent flexibility, musical recreation can also help individuals connect with the feelings and ideas of others, promoting proactive and cooperative teamwork towards shared objectives. Group singing and drumming sessions, for example, have proven effective in building camaraderie and reinforcing reality orientation among team members. These methods are also valuable in elder care facilities, encouraging interaction among residents.

Moreover, re-creative music is known to assist individuals with intellectual disabilities in developing socially acceptable and adaptive behaviors. It can also facilitate cohesiveness and cooperation within dysfunctional families or groups. In essence, "re-creating in music" refreshes the minds of both listeners and singers, offering a stimulating and enjoyable experience.

Composing

Musical composition, the creation or orchestration of a novel piece of music resulting in a musical structure, can exist as written notation or a pre-conceived acoustic event. These compositions, comprising diverse musical elements varying across individuals and cultures, can be performed from memory and involve musical notation, instrumentation, and orchestration – the art of writing music for ensembles.

Composing exercises are observed to effectively divert attention from life's harsh realities towards a more engaging and rewarding pattern-making process. Therapists and patients alike can engage in various compositional techniques, including extended techniques like alap/alapana (improvisatory exploration of a raga), improvisation itself, and musical montage (creating sound collages).

Improvisation involves spontaneous action and reaction in response to the immediate environment, while musical montage assembles sound objects or compositions from a collage of sounds. Both therapist and patient can assume the role of composer, providing an outlet for deeply felt, suppressed emotions, with the freedom to shape the composition according to their own inclinations, even incorporating "noise" if it aids emotional release.

The task of arranging or orchestrating these compositions is typically undertaken by the music therapist, sometimes with the assistance of other trained musicians. During this process, the therapist carefully considers the emotional characteristics conveyed by tone, timbre, melody, and harmony. Tailor-made musical pieces, ranging from soft to hard, pleasing to agitating, can be composed, tested, and played for the client based on their specific needs and therapeutic goals. These personalized compositions can be recorded for the individual's future use and potentially for other patients with similar issues and musical preferences.

ANAHAT TRANSFORMATIONS

Gopal Park, Behind Image Sonography Clinic,
Paud Phata Rd, Erandwane, Pune 411004

While commercially available music in the form of CDs and cassettes can be useful, tailor-made music offers the advantage of being more user-friendly and specifically aligned with a client's requirements, despite the potentially higher cost compared to off-the-shelf options. Elizabeth Swados, in her book "Listening Out Loud," emphasizes the composer's need to understand the full capabilities of each instrument, ensuring they complement rather than compete, illustrating this with an example of ineffective instrumentation. Each instrument chosen for a piece should have a clear purpose that contributes to the therapist's/composer's intended message.

Activities involving musical composition are frequently used by therapists, particularly with patients needing to develop decision-making skills and commitment. Often, the central idea of a composition is a word, thought, or feeling significant to the patient's life. For hospitalized children, songwriting can provide a means to express and understand their fears and loneliness. Composing music and writing songs can offer adults facing terminal illness an avenue to articulate profound feelings about life and death, serving as a meaningful legacy for loved ones. In de-addiction programs, group songwriting enables individuals struggling with substance abuse to examine their irrational beliefs and fears, documenting their intentions for change through song. Compositional activities can be adapted to encourage both spontaneity and planned decision-making.

Singing and/or Playing Instruments

Singing and/or playing musical instruments is a widely accepted technique in music therapy for all client groups. It is particularly beneficial for individuals with physical disabilities in developing both gross and fine motor coordination. When integrated with reading musical notation, playing instruments can aid children with learning disabilities in improving auditory-motor or visual-motor integration. For children with emotional disturbances, instrumental ensembles can be effectively used to address behavioral challenges. Certain musical instruments are also recommended for patients with intellectual disabilities, as they can facilitate a better understanding of the world around them.

In the United States, the MusicWorx team utilizes a variety of instruments, including guitar, cello, violin, keyboards, QChord, flute, and, of course, the voice for chanting, singing, and toning. Therapists also employ non-musician-proof instruments, primarily those in the percussion and rhythm categories. Additionally, CD recordings serve as a popular auditory input during therapeutic sessions.

Given that the capacity to respond to music is an inherent human trait, clients or patients do not require any prior musical experience or proficiency to achieve optimal outcomes in music therapy. As the therapy itself targets non-musical goals, individuals are not expected to possess prior music training or advanced musical skills.

Improvising

Improvisation, in its essence, is the act of spontaneous creation and response to the immediate environment. In music, this translates to composing in real-time, either just before or during a performance, occurring solo, in duets, or within groups. Often termed "extemporization," musical improvisation is the practice of creating music *ex tempore*, in the very moment it is sung or played, directly inspired by the immediate surroundings.

ANAHAT TRANSFORMATIONS

Gopal Park, Behind Image Sonography Clinic,
Paud Phata Rd, Erandwane, Pune 411004

While both music improvisation and composition are creative avenues in music therapy, improvisation is distinguished by its extemporaneous nature, unlike the more deliberate process of composition. Furthermore, improvisers generate their own music directly, whereas composers often rely on performers or orchestras to realize their musical ideas. For a music therapist, improvisation becomes a dynamic interaction where client and therapist communicate musically through spontaneous creation. The client can utilize any accessible medium, whether musical (voice, instruments like keyboard or drums, body sounds) or non-musical (images, titles, stories). As a creative process, improvisation empowers clients to explore aspects of themselves in relation to others in a positive manner, potentially enhancing their confidence in coping with challenges. It is particularly valuable for exploring the interplay between structure and freedom.

Psychologists suggest that improvisation can lead to the development of novel thought patterns, practices, structures, or symbols, as well as new ways of singing or playing, solidifying its importance in musical training. Philosophically, improvisation emphasizes bringing one's personal awareness fully "into the moment" and cultivating a deep understanding of the action being performed (e.g., singing or playing an instrument). This integration of awareness and understanding can enable practitioners to act with a wider range of appropriate responses, even in unfamiliar situations.

In music therapy settings, clients are encouraged to create songs and sing to simple melodies provided, fostering a creative mindset. This is often beneficial for individuals with mental health challenges. Beyond singing, elements of body movement (dance), comedy, and showmanship may also be incorporated, providing an outlet for emotional suppression. Across therapy sessions, improvisation has been observed to elicit gratifying emotional responses from clients.

The skills inherent in improvisation are recognized as crucial for effective communication and expression across diverse artistic, scientific, physical, cognitive, academic, and non-academic disciplines, including music. Therefore, studying improvisation skills and techniques can enhance overall efficiency. Interestingly, the mental and emotional states conducive to improvisation mirror those cultivated in spiritual and religious practices. The cycle of invention in improvisation is most effective when the practitioner possesses a thorough intuitive and/or technical grasp of the necessary skills and considerations within the improvisational domain.

Bruscia (1987) and Wigram (2004) have proposed various improvisational techniques and methods for therapeutic sessions based on improvisation principles, including:

- **Imitating:** A fundamental technique for building empathy, where the music therapist musically copies or repeats a client's response, focusing on elements like tone, sound, rhythm, interval, or even facial expression.
- **Reflecting:** A technique where the music therapist expresses the same moods or emotions displayed by the client through their improvisation.
- **Rhythmic grounding:** Establishing a steady beat or rhythm to support the client's improvisation, with a rhythmic ostinato being an example.
- **Dialoguing:** A technique where the music therapist and client communicate directly through their musical improvisations.
- **Accompanying:** The music therapist supports the client's improvisation by providing a musical accompaniment that may include rhythm, melody, and melodic or chordal progressions.

ANAHAT TRANSFORMATIONS

Gopal Park, Behind Image Sonography Clinic,
Paud Phata Rd, Erandwane, Pune 411004

Effective improvisation often involves a combination of these methods, requiring flexibility from the therapist. For instance, a therapist might integrate imitating, accompanying, dialoguing, and rhythmic grounding to maintain a dynamic session flow. According to Bruscia (1987), improvisation can be either referential, where the client improvises to represent a non-musical reference (e.g., an event, feeling, image, relationship), or non-referential, where the improvisation is based solely on the sounds of the music itself.

Some of the Variations in improvisation methods which are commonly used are explained in the table:

Method	Description	Notes
Instrumental referential	Creating tunes, melodies, or rhythms using selected musical instrument(s) guided by a specific non-musical reference (e.g., an event, feeling, image, or relationship from the client's life).	The resulting musical piece often embodies the emotions associated with the reference, potentially facilitating catharsis.
Instrumental non-referential	Creating music using selected musical instrument(s) without any external reference.	For example, providing a drum or xylophone and encouraging free, unrestricted playing.
Song improvisation	Creating original lyrics for an existing melody or singing along with a singer using a different or deviated melody.	This method is known to offer increased creative opportunities for the client.
Vocal non-referential	Creating a vocal piece spontaneously without relying on specific words, images, or pre-defined meanings.	Focus is on pure vocal sound and expression.
Body improvisation	Creating various sounds using the body with different rhythms, decibels, and frequencies.	Examples include clapping, stomping, snapping, and vocalizations without specific lyrical content.
Mixed media	Creating a musical piece using any combination of instrumental, vocal, or body sounds.	Allows for a wide range of expressive possibilities and can integrate different sensory modalities.
Conducted improvisation	Creating a musical section or piece by providing directive cues to one or more improvisers.	The therapist acts as a conductor, guiding the improvisation through verbal or non-verbal signals regarding dynamics, tempo,

ANAHAT TRANSFORMATIONS

Gopal Park, Behind Image Sonography Clinic,
Paud Phata Rd, Erandwane, Pune 411004

		articulation, and other musical elements.
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Bruscia (1998) identified several clinical goals achievable through various improvisation methods in music therapy:

1. **To establish a nonverbal channel of communication, and a bridge to verbal communication:** Improvisation can provide an initial means of connection and expression for individuals who struggle with verbal communication.
2. **To provide a fulfilling means of self-expression and identity formation:** The freedom of improvisation allows clients to explore and express their inner selves, contributing to a stronger sense of identity.
3. **To explore various aspects of self in relation to others:** Through musical interaction, clients can explore their feelings, thoughts, and behaviors in a safe and creative context with the therapist and potentially other group members.
4. **To develop the capacity for interpersonal intimacy:** Shared musical improvisation can foster a sense of connection and closeness between individuals.
5. **To develop group skills:** Group improvisational activities can enhance cooperation, turn-taking, and a sense of belonging.
6. **To develop creativity, expressive freedom, and playfulness with various degrees of structure:** Improvisation allows for exploration of creative potential within both unstructured and structured musical frameworks.
7. **To stimulate and develop the senses:** Engaging with instruments and sounds can heighten sensory awareness and integration.
8. **To play, on the spot, with a decisiveness that invites clarity of intention:** The immediate nature of improvisation encourages focused intention and action.
9. **To develop perceptual and cognitive skills:** Improvising can enhance auditory perception, motor coordination, and cognitive flexibility.

Improvisation is particularly effective in music therapy for clients needing to cultivate spontaneity, creativity, confidence, freedom of expression, communication, and interpersonal skills. While these qualities are also beneficial for engaging in improvisation itself, the process can help clients develop them. It is a recommended approach for a wide range of individuals, including those with emotional disturbances, developmentally delayed children, and adults with psychiatric disorders. Improvising facilitates communication and the sharing of feelings, while also aiding in the organization of thoughts and ideas in a meaningful way.

Beyond improvisation, music therapists utilize other methods and techniques to address the needs of individuals and groups, including:

- Music and movement/dance
- Music facilitated discussion
- Music and relaxation techniques
- Song singing

Effective application of each of these methods requires proper skills, experience, and training on the part of the music therapist to achieve desired therapeutic outcomes.

ANAHAT TRANSFORMATIONS

Gopal Park, Behind Image Sonography Clinic,
Paud Phata Rd, Erandwane, Pune 411004

Co-creative sessions between therapist and patient, focused on activating innate musicality through a variety of standard and specialized instruments, have gained popularity, especially in Western music therapy circles. By combining aesthetic sensitivity with ongoing analytical assessment, this improvisational music has aided patients in overcoming physical, emotional, and cognitive barriers. Such improvisational, creative music is used to help individuals including children with disabilities, victims of accidents and trauma, those under psychiatric or geriatric care, and self-referred adults seeking to address emotional problems and stress. This approach, known as the Nordoff-Robbins System, is increasingly popular in countries like the U.S.A., U.K., Germany, Australia, Scotland, and Japan.

Improvised Singing

Singing improvisation is an age-old art form. In some cultures, it blends musical improvisation with improvisational theatre, as was the case in Wales centuries ago. Annual competitions for poets and musicians featured improvised singing, leading to the popularity of a musical style called *penillion*. This often improvised singing was performed with harp accompaniment. The singer was required to follow the harp's freely changing melodies (*ad libitum*), striving to maintain time and conclude precisely with the harp's initiated strain. In Indian classical music concerts, the lead vocalist often initiates the musical "dialogue," which is then taken up by the accompanying instrumentalists.

An Approach to Imagination and Envisioning

Imagination is a fundamental human faculty that allows us to interpret and understand the world around us. Our senses of touch, taste, smell, sight, and hearing converge through imagination to create "images." This ability to form mental representations of sights, sounds, and feelings is crucial for problem-solving and for adopting different perspectives that can soothe and alleviate mental anxieties. Many inventions are born from imaginative inspiration. It is also considered an innate capacity to create partial or complete personal worlds within our minds, drawing from our sensory experiences of the shared reality.

In music therapy, as a musical melody unfolds, the therapist describes imaginative events, situations, and characters, which the patient then further develops. Numerous symphonies from the Western classical tradition, including those by Beethoven, Bach, Haydn, Vivaldi, and Tchaikovsky, can be utilized by therapists to stimulate the patient's imagination in response to the music played. This not only brings satisfaction to the patient but also significantly aids in overcoming issues like depression, trauma, and other psychological ailments.

The combination of music and imagery has proven effective for many therapists and researchers. It forms the basis of a music-centered psychotherapy approach where musical experiences, evoking emotions such as upheaval, calmness, compassion, and devotion, are used to facilitate necessary therapeutic changes in clients or patients. Personal experience suggests that using intense music with violent children can help them manage aggressive impulses. Similarly, very sad and melancholic songs can be helpful in processing grief or loss. Some music therapists believe that such initially coordinating music can, over a short period, be effectively replaced with musical pieces that reflect desired behavioral patterns, such as self-confidence and a joyful disposition. For example, anger can be managed by starting with intense music and gradually transitioning to relaxing, peaceful music. Likewise, sadness or depression initially addressed with somber or melancholic music can eventually

ANAHAT TRANSFORMATIONS

Gopal Park, Behind Image Sonography Clinic,
Paud Phata Rd, Erandwane, Pune 411004

be transformed into joyful tunes and beta wave-inducing music that promotes confidence and optimism.

Bonny Method of Guided Imagery and Music (GIM)

The Bonny Method of Guided Imagery and Music (GIM) represents a prominent music-assisted therapeutic modality prevalent in Western contexts. Characterized by its "inspired conception, holistic attitude, integrative intent, synergistic nature, holotropic reality, and transformational potential," this method is reported to aid patients/clients in their journey towards heightened consciousness through the engagement of auditory stimuli, visual imagery, and alternative states of awareness. Conceived and refined by Helen L. Bonny, GIM strategically combines her understanding of classical musical forms with relaxation protocols designed to elicit specific responses from individuals. The methodology involves the utilization of carefully sequenced classical music within a personalized, one-on-one session guided by a trained facilitator. Following an initial dialogue with the client, the facilitator provides relaxation and focusing instructions to facilitate entry into a relaxed state. Subsequently, an interactive music listening experience ensues, during which the client articulates the images, sensations, emotions, and awarenesses that arise in response to the music. The facilitator, concurrently attending to the music, provides support and guidance to the client's imagery experiences through various techniques. Upon the conclusion of the musical segment, the facilitator assists the client in returning from the expanded state of consciousness and facilitates the integration of the session's insights.

Drum Circles

Drum circles are informal, inclusive gatherings of drummers and percussionists of all skill levels who come together to play simple hand drums in a circle. Unlike performance-oriented drumming groups, the joy of shared rhythm and community is the focus. These circles promote equality because their structure eliminates any leadership hierarchy, ensuring everyone can freely express themselves. Welcoming participants of all ages and even simple instruments, drum circles create a vibrant and engaging atmosphere. The primary aim is to connect through rhythm, fostering a sense of unity within the group and with oneself. This experience can also build valuable teamwork skills and a strong sense of community. Introducing regular drum circles in workplaces can be a powerful tool for boosting team spirit among employees.

The facilitated drum circle

In a facilitated drum circle, a guide – not a teacher – helps the group focus and deepens the positive impact of the drumming. The focus isn't on learning to play, but on having a direct, engaging experience and lots of fun. The facilitator coordinates, guides, and encourages everyone to create exciting, on-the-spot music together, building both the group's musical flow and a sense of connection. The facilitator might start a rhythm, and then others join in, adding their own touches to build the beat naturally until it changes or fades away. Often, the circle feels relaxed, with participants feeling empowered to start and stop rhythms. The key is listening to each other, improvising freely, and knowing when to hold back. Important ideas behind drum circles are: no one is judging your playing; everyone is part of the music; the music you make in the moment is it – no

ANAHAT TRANSFORMATIONS

Gopal Park, Behind Image Sonography Clinic,
Paud Phata Rd, Erandwane, Pune 411004

rehearsals needed; there's no right or wrong way to play; and the drum circle is a safe space to try things out.

A recent study in a care home in southern Ontario, Canada, looked closely at how residents and day program attendees felt about being part of a drum circle. Researchers watched 20 sessions and talked to participants and facilitators about their experiences. They found four main ways people were affected: feeling more powerful, expressing themselves more, feeling closer to others, and a sense of community. The study showed that playing instruments with rhythm was a good experience for most people, helping them communicate better, feel a sense of accomplishment, and enjoy making music together.

Use of Musical Vibrations

Music therapy can be particularly beneficial for the hearing-impaired by leveraging their other senses, especially the sense of touch, to engage with music. Here's how:

- **Vibrotactile Instruments:** As highlighted in the text, vibrotactile instruments play a crucial role. These instruments allow individuals with hearing loss to perceive and interpret sounds through vibrations felt on their skin. This provides a tangible way to experience the physical aspects of music, such as rhythm, tempo, and even melodic contours to some extent.
- **Sensory Substitution:** Music therapy utilizes the principle of sensory substitution, where one sense (touch) compensates for the reduced function of another (hearing). By focusing on the tactile sensations produced by music, therapists can create meaningful musical experiences for individuals who cannot fully access auditory information.
- **Enhanced Communication:** The research mentioned in the text indicates that vibrotactile instrumental rhythm playing can enhance communication for the hearing-impaired. Feeling the vibrations of music can provide a shared sensory experience that fosters connection and interaction, even without relying on spoken language or traditional auditory cues.
- **Promotion of Communal Music-Making:** Engaging with vibrotactile instruments in a group setting, such as a drum circle, allows for collective music-making. Participants can feel the shared rhythms and contribute their own, fostering a sense of community and belonging, as the Canadian study observed.
- **Emotional Expression and Engagement:** Music, regardless of how it's perceived, can be a powerful tool for emotional expression. For the hearing-impaired, feeling the vibrations can evoke emotional responses and provide a non-verbal outlet for feelings. The "musical engagement" can be strong through vibrotactile stimulation, encompassing a sense of being part of the music, arousal, and groove.
- **Cognitive Stimulation:** Engaging with music through touch can still provide cognitive stimulation, such as the recognition of rhythmic patterns and the anticipation of changes in the music.
- **Physical and Motor Skills:** Playing vibrotactile instruments can also promote physical and motor skills, such as coordination and fine motor control, depending on the instrument.

The Canadian research study underscores the positive impact of vibrotactile instrumental rhythm playing on hearing-impaired individuals in a long-term care facility. The identified behavioral categories of empowerment, bringing out, bringing together, and community resonance further highlight the holistic benefits of this approach within music therapy. By focusing on the sense of touch, music therapy can unlock the power of music for those with hearing impairments, fostering communication, emotional well-being, and social connection.

ANAHAT TRANSFORMATIONS

Gopal Park, Behind Image Sonography Clinic,
Paud Phata Rd, Erandwane, Pune 411004

Vibroacoustics

Sound originates from vibrating objects. Whether it's the stretched string of a guitar or veena, the vibrating surface of a drum or cymbal, or the air column within a pipe or didgeridoo, these vibrations produce sound. A common acoustic phenomenon also demonstrates how air moving rapidly past an obstruction can create turbulence, as seen when blowing a whistle.

Musical pitch is determined by the frequency of these vibrations, measured in Hertz (Hz). One Hz signifies one vibration per second. Typically, vibrations above 100 Hz are considered "high frequency," while those below are "low frequency." The human auditory range generally falls between 20 Hz and 20,000 Hz.

The term 'vibroacoustics' describes the process of perceiving sound vibrations through the body. Some therapists utilize specialized tuning forks applied to specific meridian points on a client's body to achieve a therapeutic effect. This practice later evolved into the use of specially designed furniture, such as chairs, tables, and beds, all incorporating powerful speakers. The intended outcomes of these vibroacoustic designs include a significant physical and physiological impact by enabling the body to oscillate in response to the vibrations.

Vibroacoustic Therapy

Inspired by a deaf-mute patient, physician Teirich (1959) conducted a study on music and vibration. He observed that his 59-year-old patient found a unique way to experience music, explaining how his back served as the primary "receiving station." The patient had developed an "inner sense of vibration" that allowed him to perceive sound waves. Consequently, Teirich constructed a couch equipped with loudspeakers that could transmit the vibrations of J.S. Bach's D minor Toccata and Fugue directly to the patient's solar plexus. He also used fellow physicians as a control group, who reported sensations of immediate warmth in the solar plexus, complete stomach relaxation, and a pleasant, dreamlike state.

Teirich's groundbreaking work in developing vibroacoustic (VA) therapy paved the way for researchers like Olav Skille (1982), a Norwegian music educator, who further advanced the practice.

Vibroacoustics involves using sinusoidal, low-frequency sound pressure waves between 30-120 Hz combined with music for therapeutic purposes. Skille's interest in the physical effects of sound vibration on children with intellectual disabilities led him to investigate whether it could induce relaxation in individuals with severe physical and mental disabilities and reduce muscle tone. He promoted relaxation by playing music through loudspeakers placed against beanbags on which the children lay. His work eventually led to the Norwegian company VibroAcoustics AS manufacturing and marketing the VibroAcoustic chair (VA100) and vibroAcoustic bed (VB500) in the late 1980s.

Vibroacoustic Chair and Bed

The VibroAcoustic (VA) chair (VA100) and bed (VB500) were designed using stereophonic tapes to deliver a specific musical form: a single, low-frequency, rhythmically pulsed sinusoidal sound wave intended to promote relaxation. This pulsed effect was achieved by combining two closely spaced sinusoidal tones, for instance, 40 Hz and 40.5 Hz.

Within the chair, six loudspeakers (20-1500Hz) strategically positioned applied this vibratory stimulus to different body areas. A control unit (SU200) managed three key functions: adjusting the stimulus intensity for each body region, balancing the music and pulsed tones (allowing for music-only listening), and controlling the headphone volume for the music component.

ANAHAT TRANSFORMATIONS

Gopal Park, Behind Image Sonography Clinic,
Paud Phata Rd, Erandwane, Pune 411004

These devices could generate physical vibrations felt by the body when connected to a sound source. The varying vibrational frequencies within music allowed for simultaneous vibrations at different speeds. The underlying principle was that these soothing vibrations could relax muscles and enhance blood circulation as sound waves traveled throughout the body.

In VA therapy, music is delivered through loudspeakers integrated into pads, beds, recliners, tables, chairs, etc., resulting in the music being perceived both auditorily and tactilely. This forms the theoretical basis for using selected music as both an auditory and a vibratory stimulus in VA therapy, aiming to elicit a relaxation response in patients. Its non-invasive nature and the therapist's confidence in its pleasing effects have contributed to its increasing popularity in hospitals and care facilities.

VA therapy typically employs audible frequencies, based on the belief that soft, low-frequency, non-rhythmic music can act as a sedative input, potentially inducing physiological changes conducive to relaxation. Research suggests that such "sedative" music does encourage relaxation.

The VA therapeutic concept acknowledges the physical effects of sound, supported by the growing recognition that individuals are sensitive to varying levels of mechanical vibration applied to the body. Research on skin temperature changes in relation to vibration amplitude further supports this. These scientific observations, combined with numerous subjective positive experiences, have reinforced the relaxing role of "sedative" music and individuals' sensitivity to vibration.

Vibro-acoustic therapy is essentially a "passive" intervention involving the transmission of pre-recorded music into the patient's body via speakers embedded in furniture. The book "Music Vibration and Health" (1997) serves as a valuable resource on VA therapy, documenting clinical and research studies, treatment procedures, ethical considerations, and contraindications.

VA therapy has shown benefits for patients with cystic fibrosis (potentially loosening lung mucus) and in reducing the severity of asthma attacks. Studies with cardiac patients suggest that Physioacoustic therapy could improve cardiac input, potentially reducing the need for increased heart support medication.

Skille (1992) reported positive outcomes when incorporating Vibroacoustics into rehabilitation programs for hearing-impaired individuals. Research by Darrow and Goll (1989) and Darrow (1992) indicated that the Somatron improved rhythm identification and pitch discrimination in these patients.

Comparative research between VibroAcoustics and music intervention showed that VA therapy had a greater impact on anxiety in individuals with intellectual disabilities and improved tolerance of staff interactions in those with self-injurious behavior. It's proposed that the pleasurable tactile experience of VA therapy can counteract the self-stimulatory aspect of unprovoked self-injury and reduce anxiety in acutely ill children during procedures and encourage interaction in autistic patients.

VA therapy has also been suggested as a potential remedy for stress-related conditions, although these findings warrant cautious interpretation due to the lack of double-blind trials, raising the possibility of positive expectation bias.

Anecdotal evidence and objective studies suggest that VA therapy is pleasurable and relaxing, making it potentially useful for addressing anxiety and related problems.

ANAHAT TRANSFORMATIONS

Gopal Park, Behind Image Sonography Clinic,
Paud Phata Rd, Erandwane, Pune 411004

Ongoing clinical research explores VA therapy's effects on pain relief. Some findings suggest it may reduce pain perception in rheumatoid arthritis and fibromyalgia sufferers, as well as post-operative and injury-related pain.

Olav Skille's initial belief that VibroAcoustics could relax individuals and reduce muscle tone has been supported by research. Studies comparing VA therapy to music alone in cerebral palsy patients showed greater reductions in muscle tone and improved range of motion. VA therapy is seen as a potential long-term preventive and treatment intervention for reducing the risk of fixed deformities in certain individuals. Case studies and work with Rett Syndrome have provided further evidence of this response.

Anecdotal reports also suggest that VibroAcoustics can provide relief from pain associated with conditions like polyarthritis, rheumatism, and colic.

Music Vibration Table (MVT)

The Music Vibration Table (MVT) is a patented system under development at the Texas Centre for Music and Medicine in the USA. Built upon a standard hospital procedural stretcher, it comprises a base table, a sound system, a computerized vibration feedback processing system, and a vibrating tabletop membrane. Padding is placed between the membrane and the patient to enhance comfort during treatment.

The vibrating membrane features three independently controlled modules positioned beneath the torso, lower legs, and thighs. Within each module, music selections are processed by a separate computerized vibration system. This system regulates the magnitude and amplitude of vibrations within the 60-300 Hz range and records the parameters and timing of stimulus variations. External sound is provided through speakers connected to the system, with the volume adjustable independently of the vibration levels to suit each patient's preference.

Chesky and his team (1996) specifically designed the MVT for pain relief. They proposed a dual-pronged theory for its effectiveness, combining the attention-distraction and emotional (affective) aspects of music (Brown, Chen, & Dworkin, 1991) with the physiological effects of vibration. They argued that MVT could be administered proactively before pain onset as well as reactively. Consequently, they hypothesized that MVT had the potential to lessen the need for pre-operative sedatives, reduce post-operative pain, and shorten hospital stays.

Somatron

In 1985, Byron Eakin, a professional artist, conducted an experiment by connecting a pair of loudspeakers to a massage table to explore relaxation techniques. This seemingly simple setup is now considered a significant historical point, leading to the development of a range of commercial products known as Somatron products.

These products include the Clinical Recliner, which features four speakers: two at ear level for auditory input and two 40-8000Hz vibroacoustic speakers located in the lower leg and back sections. These vibroacoustic speakers are integrated into an 11"x14" frame, referred to as a "vibroacoustic diaphragm," designed to enhance the transmission of vibrations from specially composed vibroacoustic music.

In the United States, Somatron products are utilized by music therapists and other healthcare professionals, including masseurs, spa owners, and lifestyle gurus. Several research reports published in the Journal of Music Therapy have investigated the therapeutic effects of Somatron technology.

ANAHAT TRANSFORMATIONS

Gopal Park, Behind Image Sonography Clinic,
Paud Phata Rd, Erandwane, Pune 411004

One study conducted in 1991 at Florida State University compared the experience of college students listening to music and the sound of a dentist's drill via both a tape player and a Somatron. The findings indicated that regardless of the sound type, vibrotactile stimulation appeared to "blunt perception," as students reported a reduction in both their comfort and discomfort zones. When the sounds were experienced through the Somatron, students were more likely to exhibit an increase in finger temperature, a physiological indicator of improved relaxation. These temperature increases were statistically significant ($p < 0.05$). Additionally, students using the Somatron tended to show a decrease in heart rate after the first sound presentation (whether music or drill) followed by an increase after the second sound, a pattern opposite to that observed in students listening via a tape player. Interestingly, students using the Somatron generally reported liking the experience, with some even preferring the sound of the dental drill when experienced this way. Males also showed a greater preference for the Somatron experience compared to females.

Another unpublished study from Miami investigated the effects of a relaxation session using a Somatron on muscle tension, measured by electromyographic equipment (EMG). Students were divided into three groups: one listened to "soothing music," another to a pre-recorded muscle relaxation program, and the third to both simultaneously, all via the Somatron. The researcher found a tendency for increased muscle tension in the group listening to soothing music alone, while the other two groups showed a tendency for reduced muscle tension. However, these results were not statistically significant and somewhat contradicted the students' verbal reports. Students listening to soothing music reported feeling relaxed despite increased muscle tension, while those using the relaxation program reported feeling increased tension despite reduced muscle tension. The researcher suggested a small sample size and uncontrolled environmental distractions as potential reasons for these discrepancies.

An additional unpublished research paper from St Mary's Hospital in West Palm Beach, Florida, compared the effects of music presented vibrotactily via a Somatron recliner versus non-vibrotactily via a regular reclining chair on anxiety levels in psychiatric patients and staff. Both groups experienced a statistically significant reduction in anxiety following fifteen minutes of music. Notably, the reduction in anxiety was reported to be significantly greater in the group using the Somatron compared to the group listening to music via a cassette player.

However, a study by Hooper and Lindsay (1997) reported a significant difference in anxiety reduction response to Somatron treatment in two individuals with learning disabilities. Furthermore, several comparative studies have found no additional benefit to VA therapy over non-vibroacoustic interventions.

In summary, Byron Eakin's early experiment connecting loudspeakers to a massage table marked the beginning of Somatron products and the exploration of vibroacoustic therapy. While some research suggests positive effects like blunted perception of unpleasant sounds, increased relaxation indicators, and greater anxiety reduction compared to auditory music alone, other studies show inconsistent results and question the added benefits of vibroacoustics over traditional music therapy.

The Vibro-acoustic Devices: A Summary

The following table summarizes the nature and description of the Vibroacoustic devices, we have just discussed in the foregoing paragraphs.

Name	Vibro Acoustics	Physioacoustic	Somatron	Music
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		therapy		Vibration Table (MVT)
Inventor/ Developer	Olav Skille (Norway)	Petri Lehikoinen (Finland)	Byron Eakin (America)	Kris Chesky (America)
Product	VibroAcoustic Chair (VA100)	Physioacoustic mattress	Clinical Recliner	MVT
Details/ Components	-chair -6 speakers (20-1500 Hz) -control unit -audio system	-mattress -4 transducers (27-113 Hz) -computer unit -audio system	-chair -4 speakers (40-8000 Hz) - vibroacoustic diaphragm -audio system	-base table -vibrating membrane (60-300 Hz) -vibration processing system -audio System
Sound/music Stimulus	-relaxing music - single low frequency rhythmical-ly pulsed sinusoidal sound wave	-specially selected or composed music.	- vibroacoustic music.	-music chosen by the researcher.
Availability	Not available	Supplier: Heritage Medical Products, Kalamazoo, MI, USA.	Supplier: Somatron Corporation, Tampa, FL, USA.	Patented system.
Remarks	Combines music as well as the sinusoidal sound.	Combines music as well as the sinusoidal sound.	A mattress through which music is felt and experienced as vibrations	Provides a measure of the vibration experienced by a subject.

VA Therapy: Contraindicated Conditions

Visual-auditory (VA) therapy should not be used in individuals with psychosis, acute inflammation, pregnancy, haemorrhaging, thrombosis, or hypertension, nor in those with pacemakers. Currently, however, VA equipment is available for purchase and use by anyone. Expressing their concern regarding this lack of regulation, Wigram and Dileo (1997) proposed minimum standards for its application in private or institutional settings. They recommended that practitioners possess professional certification and/or training in a healthcare field, along with specialized training in the use of music for VA therapy. Furthermore, Wigram and Dileo (1997) suggested that qualified music therapists may be competent to administer this form of treatment.

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Is VA Therapy a form of Music Therapy ?

Music now frequently serves as a complementary or supplementary treatment. Many healthcare facilities worldwide routinely play background music in wards and waiting rooms after obtaining patient consent. While such music may offer therapeutic benefits, it is not classified as music therapy.

Maranto (1993) specified situations where the use of music in medicine can be considered music therapy. In these instances, music addresses specific, often individualized, therapeutic goals by either supporting medical treatment or acting as an adjunct to a primary intervention for a particular medical condition.

As we've seen, visual-auditory (VA) therapy is a distinct treatment approach compared to simply playing therapeutic music. American music therapists appear to be more comfortable with this method than their European counterparts. Because the presentation of music in VA therapy often serves as a relaxant (or a contingent reinforcer), the resulting therapeutic process may not involve a deeply dynamic and evolving musical relationship. In contrast, Europe has a history of active music making within music therapy.

Consistent with this trend, recent studies demonstrate the application of VA therapy to support the medical treatment of premature infants (Burke et al., 1995), as well as gynaecological (Burke, 1997), cardiac (Butler & Butler, 1997), and post-operative patients (Burke & Thomas, 1997). VA therapy has also been used as a standalone treatment for high muscle tone (Wigram, 1997b), pain relief (Chesky, 1992), and anxiety (Jones, 1997). In these cases, VA therapy was employed with a focus on achieving specific outcomes.

A VA therapy session involves familiarizing the patient with the equipment (bed, mattress, etc.) and ensuring their comfort by adjusting stimulus levels. Patient responses are monitored discreetly. At the session's conclusion, the patient receives reassurance, guidance, and support, allowing the therapist to establish a therapeutic relationship. This process requires the skills of a music therapist not only for music selection but also, particularly with non-verbal clients, for understanding and interpreting individual responses to the stimuli.

VA therapy provides a unique auditory experience. It systematically applies the fundamental principle that low frequencies have a relaxing effect within the context of a therapeutic relationship. Therefore, VA therapy can be considered a modified form of music therapy.

Physioacoustic Therapy

In the 1970s, Finland saw the initiation of experiments on physioacoustic therapy, a slightly adapted version of VA therapy.

While VibroAcoustics merges a single frequency sinusoidal sound with music, the Physioacoustic system generates and combines a spectrum of sinusoidal frequencies with the musical stimulus.

The Physioacoustic system comprises a mattress, a computer unit, an audio system, and a power source. Four transducers are embedded within the mattress, positioned to be under the neck, lower back, thigh, and lower leg areas. This system integrates low-frequency sinusoidal sound with specifically chosen music played through the audio system. The computer generates the sinusoidal sound and controls its frequency, pulsation, and direction. It's programmed to vary the sound

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Paud Phata Rd, Erandwane, Pune 411004

between 27Hz and 113Hz until the optimal resonance for a muscle group is achieved, eliciting a response. The sound is pulsed to allow for rest periods and prevent the numbness often associated with continuous stimulation. The sound's direction can be adjusted to move from the lower body upwards or vice versa. Furthermore, the sounds and programs can be modified to suit individual client needs, such as general relaxation, sleep induction, or pulmonary physiotherapy.

Lehikoinen (1997) documented a positive response to Physioacoustic therapy in a patient with a genetic brain disease. Research also indicates that the 40Hz brain wave diminishes or is disrupted in the early stages of Alzheimer's disease and certain brain injuries (Llinas & Ribari, 1993). Based on its use of the 40Hz frequency, Lehikoinen proposed that Physioacoustic therapy might reinforce this thalamic frequency, potentially offering an intervention for patients with brain injuries and stroke.

Vibroacoustic Music Therapy (VAM) for Pain and Symptom Reduction in Chemotherapy

In the year 2000, the Ella Milbank Foshay Cancer Center at Jupiter Medical Centre in Florida (USA) initiated a Vibroacoustic Music (VAM) therapy program. The program's objective was to alleviate pain and symptoms experienced by patients due to cancer and its treatment. It focused on administering VAM to outpatient chemotherapy and radiation therapy patients, as well as cancer survivors. This service was provided free of charge, and patients could attend as many sessions as they wished. The outcomes of the vibroacoustic sessions were monitored during a 5-week program evaluation. The results of this Foshay evaluation, encompassing 41 sessions, indicated a 61% to 74% reduction in pain and symptoms among the cancer patients. The program remains active, and data collection for ongoing evaluation continues.

Binaural Beat Frequencies (BBFs)

Recent years have marked significant advancements in psychoacoustic sound technology.

The emergence of binaural beat frequencies (BBFs) has introduced a potent auditory tool for self-improvement. The perception of these beats originates within the brain, independent of external physical stimuli.

Binaural beat frequencies are created by embedding two slightly detuned tones beneath music, nature sounds, or white noise soundtracks. When listened to in isolation, binaural beats can sound somewhat like a motorboat. Consequently, they are typically mixed subtly into the soundtrack, making their presence nearly imperceptible to the ears. However, the brain tracks these frequencies through a process called entrainment. These precise tones can gently accelerate or decelerate brainwave activity, functioning either as an auditory stimulant or as a sleep aid.

While all four brainwave states (beta, alpha, theta, and delta) are simultaneously present in the brain, it's reported that binaural beats can be used to amplify one of these states, making it dominant. This suggests that the qualities associated with each brainwave state, as outlined below, can be enhanced through BBFs:

- **Beta:** Concentration and Learning

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- **Alpha:** Relaxation and Stress Reduction
- **Theta:** Healing
- **Delta:** Sleep

It's also reported that binaural beats can influence brain functions beyond those directly related to hearing. This phenomenon is known as the frequency following response. The underlying principle is that if the brain receives a stimulus with a frequency within the range of brainwaves, the dominant brainwave frequency is likely to shift towards the frequency of the stimulus (entrainment). This stimulus can be auditory, visual, or a combination of both (the Dreamachine is an example of this). Binaural beat stimulation has been used quite extensively to induce various states of consciousness.

In recent years, binaural beats have garnered significant interest, particularly among neurophysiologists investigating the sense of hearing. Given their potential to influence the brain in subtle ways through brainwave entrainment, binaural beats may offer benefits for relaxation and overall well-being. However, it is advisable for individuals who frequently drive or operate heavy machinery to avoid binaural beat frequencies. Similarly, those with a predisposition to seizures should also refrain from using them.

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